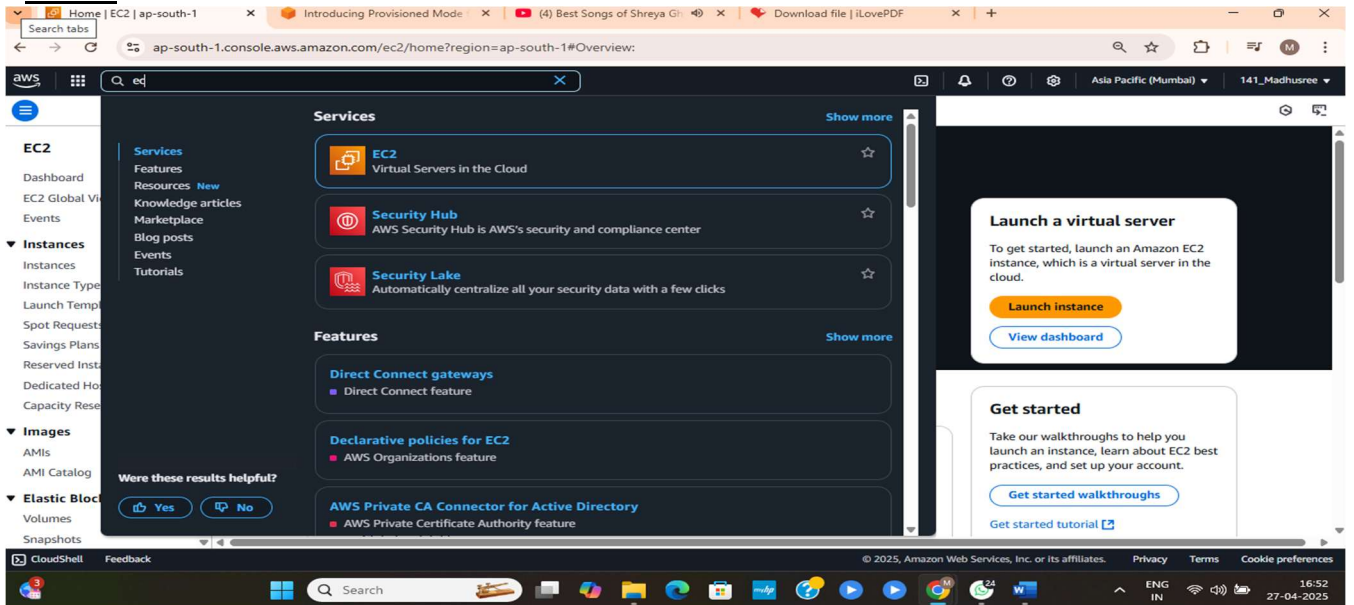


ASSIGNMENT - 14

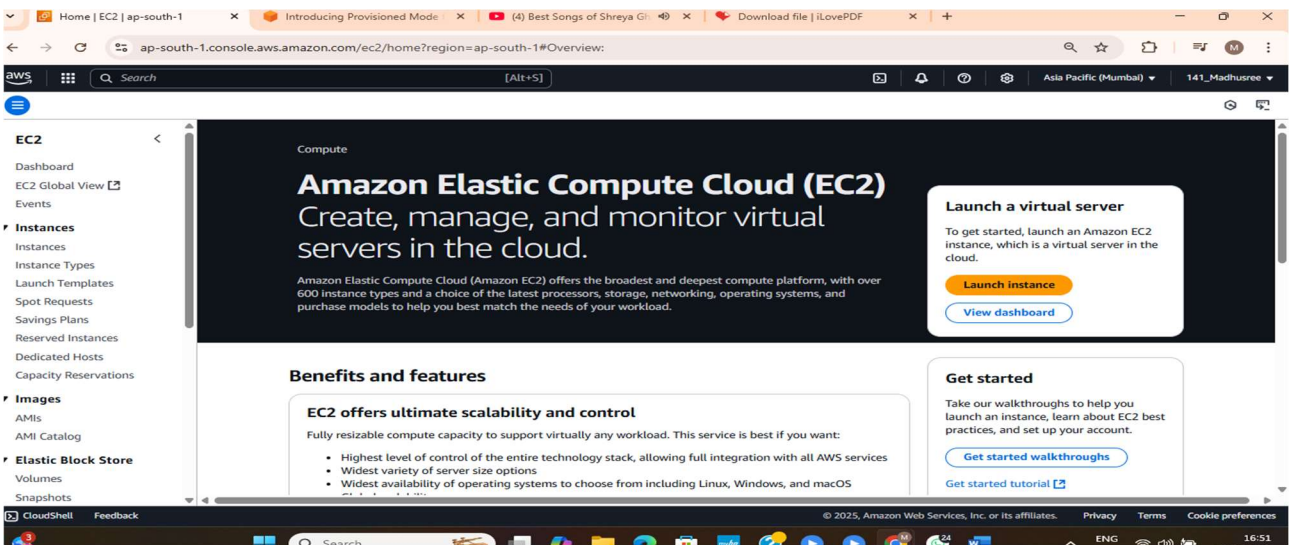
PROBLEM STATEMENT:

Create an Elastic IP for an Instance.

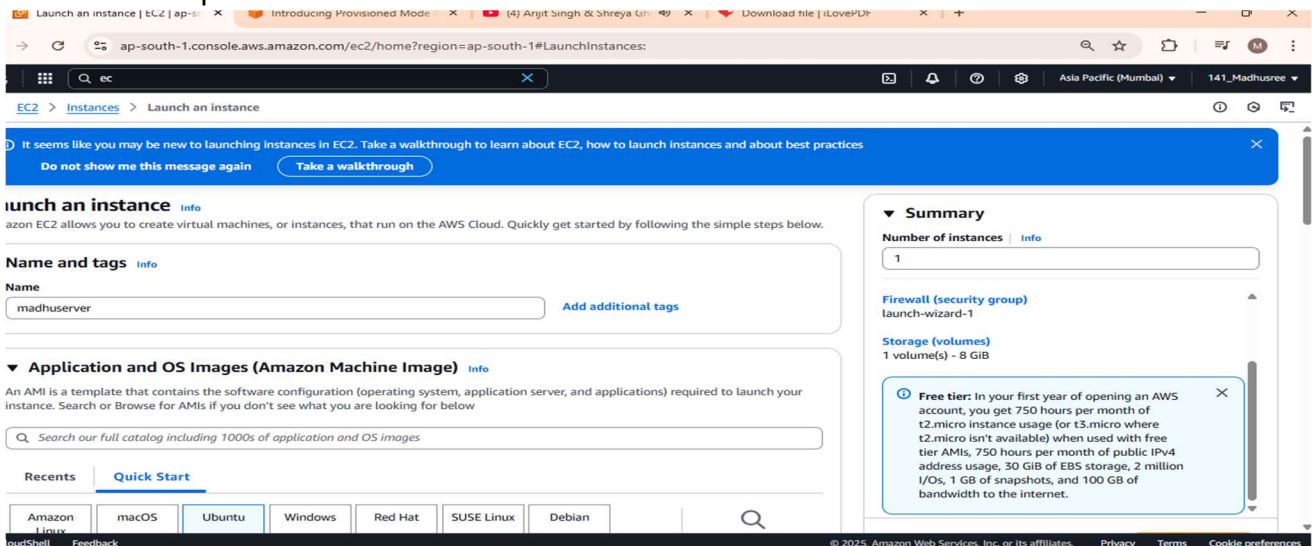
STEP 1-> Go to “EC2” from the AWS homescreen.



STEP 2-> From the left side menu, go to “Instances”.



STEP 3-> Select an instance or Launch a new one and save its Public IPv4 address in a text file for future comparison.



Launch an instance | EC2 | ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

Key pair name - required
 madhukey [Create new key pair](#)

Network settings [Info](#) [Edit](#)

Network [Info](#)
 vpc-040f39b77897466c2

Subnet [Info](#)
 No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)
 Enable
 Additional charges apply when outside of free tier allowance

Firewall (security groups) [Info](#)
 A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)
 Select security groups
 launch-wizard-1 sg-01d470503aecb044e [Compare security group rules](#)

Summary

Number of instances [Info](#)
 1

Firewall (security group)
 launch-wizard-1

Storage (volumes)
 1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

[Cancel](#) [Launch instance](#) [Preview code](#)

Launch an instance | EC2 | ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#InstanceDetails:instanceid=i-0b0ffb55a3c1ea32

Instance summary for i-0b0ffb55a3c1ea32 (madhuserver) [Info](#) [Connect](#) [Instance state](#) [Actions](#)

Updated less than a minute ago

Instance ID i-0b0ffb55a3c1ea32 IPv6 address - Hostname type IP name: ip-172-31-11-229.ap-south-1.compute.internal Answer private resource DNS name IPv4 (A) Auto-assigned IP address 3.110.189.129 [Public IP] IAM Role - IMDSv2 Required	Public IPv4 address 3.110.189.129 open address Instance state Running Private IP DNS name (IPv4 only) ip-172-31-11-229.ap-south-1.compute.internal Instance type t2.micro VPC ID vpc-040f39b77897466c2 Subnet ID subnet-0cccd921cebe69fbb Instance ARN arn:aws:ec2:ap-south-1:491085397121:instance/i-0b0ffb55a3c1ea32	Private IPv4 addresses 172.31.11.229 Public IPv4 DNS ec2-3-110-189-129.ap-south-1.compute.amazonaws.com open address Elastic IP addresses - AWS Compute Optimizer finding - Auto Scaling Group name - Managed false
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STEP 4-> With the instance selected, open the Instance State dropdown menu and select **"Stop Instance"** option to temporarily put the instance in the sleep state. Now again start this instance.

Dashboard
 EC2 Global View
 Events

Instances
 Instances
 Instance Types
 Launch Templates

Instances (1/1) [Info](#) Last updated 1 minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) All states

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
☒	madhuserver	i-0b0ffb55a3c1ea32	Stopped	t2.micro	Initializing	View alarms	ap-south-1b	-

EC2 [Dashboard](#) [EC2 Global View](#) [Events](#)

Instances
 Instances
 Instance Types
 Launch Templates

Notifications 0 0 2 0 0 0

Instances (1/1) [Info](#) Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) All states

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
☒	madhuserver	i-0b0ffb55a3c1ea32	Running	t2.micro	Initializing	View alarms	ap-south-1b	ec2-3-110-189-129.ap-south-1.compute.amazonaws.com

STEP 5-> Now again save the Public IPv4 address in the text file and compare it with the previous IP.

Instance summary for i-0b0ffbf55a3c1ea32 (madhuserver) Info

Updated less than a minute ago

Instance ID: i-0b0ffbf55a3c1ea32

IPv6 address: -

Hostname type: IP name: ip-172-31-11-229.ap-south-1.compute.internal

Answer private resource DNS name: IPv4 (A)

Auto-assigned IP address: 3.110.164.253 [Public IP]

IAM Role: -

Public IPv4 address: 3.110.164.253 | open address

Instance state: Running

Private IP DNS name (IPv4 only): ip-172-31-11-229.ap-south-1.compute.internal

Instance type: t2.micro

VPC ID: vpc-040f39b77897466c2

Subnet ID: subnet-0cccd921cebe69fbb

Instance ARN: arn:aws:ec2:ap-south-1:491085397121:instance/i-0b0ffbf55a3c1ea32

Private IPv4 addresses: 172.31.11.229

Public IPv4 DNS: ec2-3-110-164-253.ap-south-1.compute.amazonaws.com | open address

Elastic IP addresses: -

AWS Compute Optimizer finding: Opt-in to AWS Compute Optimizer for recommendation | Learn more

Auto Scaling Group name: -

Managed: false

Before assigning Elastic IP

3.110.189.129

3.110.164.253

It is evident that the IP address changes every time the instance is launched. To avoid it Elastic IP has to be assigned.

STEP 6-> From the EC2 Dashboard, go to the “Elastic Ips” option click on “Allocate Elastic IP address”

Elastic IP addresses

Find elastic IP addresses by attribute or tag

Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record
No Elastic IP addresses found in this Region				

View IP address usage and recommendations to release unused IPs with Public IP insights

Allocate Elastic IP address

STEP 7-> With “Amazon’s pool of IPv4 Addresses” selected under Public IPv4 Address pool, click on “Allocate”.

Allocate Elastic IP address Info

Elastic IP address settings Info

Public IPv4 address pool

- ☒ Amazon's pool of IPv4 addresses
- ☐ Public IPv4 address that you bring to your AWS account with BYOIP. (option disabled because no pools found) Learn more
- ☐ Customer-owned pool of IPv4 addresses created from your on-premises network for use with an Outpost. (option disabled because no customer owned pools found) Learn more
- ☐ Allocate using an IPv4 IPAM pool (option disabled because no public IPv4 IPAM pools with AWS service as EC2 were found)

Network border group: ap-south-1

Global static IP addresses

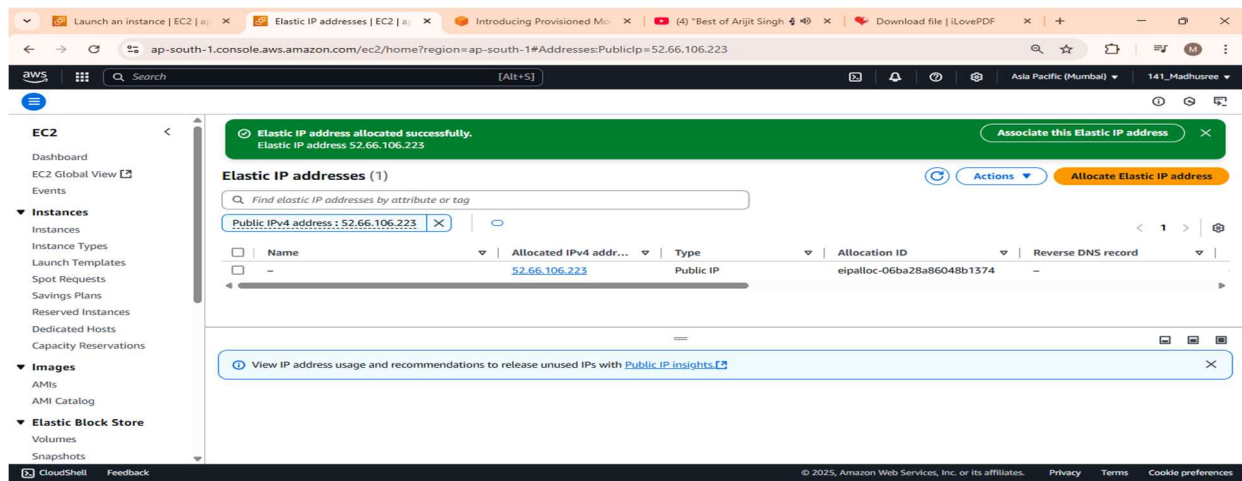
AWS Global Accelerator can provide global static IP addresses that are announced worldwide using anycast from AWS edge locations. This can help improve the availability and latency for your user traffic by using the Amazon global network. Learn more

Create accelerator

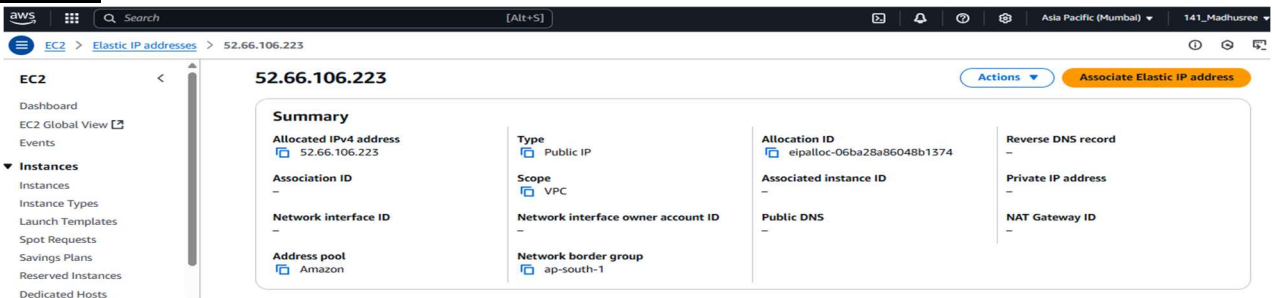
Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

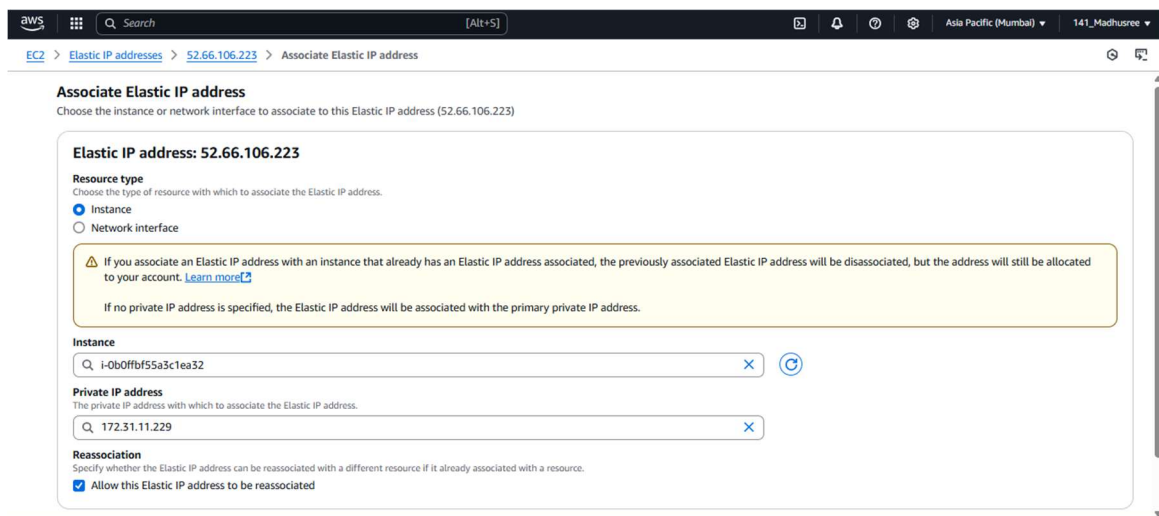
STEP 8-> The Elastic IP has been created. Click on the IP to open it.



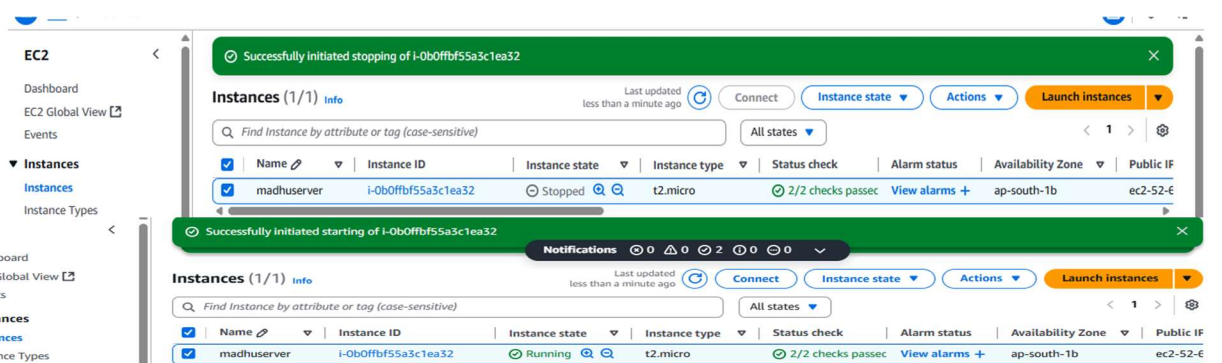
STEP 9-> Click on “Associate Elastic IP Address”.



STEP 11-> With resource type selected as “Instance”, select the name of the instance & its private IP address. Check the bottom checkbox to reassociate the Elastic IP. Then click on “Associate”.



STEP 12-> Copy the IPv4 address before selecting “Stop Instance” and again start this instance and copy the IPv4 address.



STEP 13-> No changes are observed.

